

§7-3 Trigonometric Substitution

Homework : 5,9,11,17,21,27,30,34,41

題型：(利用三角代換將根號除掉)

$$(I) \sqrt{a^2 - x^2} \xrightarrow{x=a \sin \theta} a \cos \theta, \theta \in I \cup IV.$$

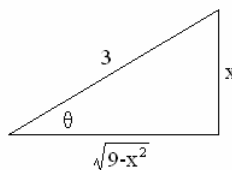
$$(II) \sqrt{x^2 - a^2} \xrightarrow{x=a \sec \theta} a \tan \theta, \theta \in I \cup III.$$

$$(III) \sqrt{x^2 + a^2} \xrightarrow{x=a \tan \theta} a \sec \theta, \theta \in I \cup IV.$$

Example 1 : $\int \frac{\sqrt{9-x^2}}{x^2} dx$.

Solution :

$$\begin{aligned} & \int \frac{\sqrt{9-x^2}}{x^2} dx & x = 3 \sin \theta & \Rightarrow dx = 3 \cos \theta d\theta \\ & = \int \frac{3 \cos \theta}{9 \sin^2 \theta} 3 \cos \theta d\theta \\ & = \int \frac{1 - \sin^2 \theta}{\sin^2 \theta} d\theta \\ & = -\cot \theta - \theta + C \\ & = -\frac{\sqrt{9-x^2}}{x} - \sin^{-1} \frac{x}{3} + C. \end{aligned}$$

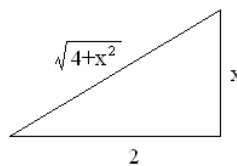


Example 2 : $\int \frac{dx}{x^2 \sqrt{x^2 + 4}}$.

Solution :

$$\begin{aligned} & \int \frac{dx}{x^2 \sqrt{x^2 + 4}} & x = 2 \tan \theta & \Rightarrow dx = 2 \sec^2 \theta d\theta \\ & = \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta 2 \sec \theta} \\ & = \frac{1}{4} \int \frac{\sec \theta d\theta}{\tan^2 \theta} \\ & = \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \int \frac{1}{u^2} du \quad (u = \sin \theta) \\
&= -\frac{1}{4} \frac{1}{u} + C \\
&= -\frac{1}{4} \frac{1}{\sin \theta} + C \\
&= -\frac{1}{4} \frac{\sqrt{x^2 + 4}}{x} + C.
\end{aligned}$$



Example 3 : $\int \frac{x}{\sqrt{x^2 + 4}} dx = (x^2 + 4)^{\frac{1}{2}} + C.$

Solution :

$$\begin{aligned}
&\int \frac{x}{\sqrt{x^2 + 4}} dx \\
&= (x^2 + 4)^{\frac{1}{2}} + C.
\end{aligned}$$

註：若 Substitution 的方法可解決，優先使用。

Example 4 : $\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2 + 9)^{\frac{3}{2}}} dx.$

Solution :

$$\begin{aligned}
&\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2 + 9)^{\frac{3}{2}}} dx \quad x = \frac{3}{2} \tan \theta \quad \Rightarrow \quad dx = \frac{3}{2} \sec^2 \theta d\theta \\
&= \int_0^{\frac{\pi}{3}} \frac{\frac{27}{8} \tan^3 \theta \cdot \frac{3}{2} \sec^2 \theta}{27 \sec^3 \theta} d\theta \\
&= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\tan^3 \theta}{\sec \theta} d\theta \\
&= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta \\
&= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{(1 - \cos^2 \theta)}{\cos^2 \theta} \sin \theta d\theta
\end{aligned}$$

$$\begin{aligned}
&= \frac{-3}{16} \int_{\frac{1}{2}}^1 \frac{1-u^2}{u^2} du \quad (u = \cos \theta) \\
&= \frac{3}{16} \int_{\frac{1}{2}}^1 (u^{-2} - 1) du \\
&= \frac{3}{16} \left(-u^{-1} - u \Big|_{\frac{1}{2}}^1 \right) \\
&= \frac{3}{32}.
\end{aligned}$$

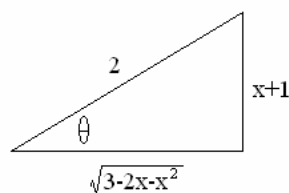
Example 5 : $\int \frac{x \, dx}{\sqrt{3-2x-x^2}}.$

Solution :

$$\begin{aligned}
&\int \frac{x \, dx}{\sqrt{3-2x-x^2}} \\
&= \int \frac{x \, dx}{\sqrt{4-(x+1)^2}} \\
&= \int \frac{-1+2\sin\theta}{2\cos\theta} 2\cos\theta d\theta \\
&= -\theta - 2\cos\theta + C \\
&= -\sin^{-1} \frac{x+1}{2} - \sqrt{3-2x-x^2} + C.
\end{aligned}$$

$$x = -1 + 2\sin\theta$$

$$dx = 2\cos\theta d\theta$$



Example 6 : $\int_0^{\frac{\pi}{2}} \frac{\cos t}{\sqrt{1+\sin^2 t}} \, dt.$

Solution :

$$\begin{aligned}
&\int_0^{\frac{\pi}{2}} \frac{\cos t}{\sqrt{1+\sin^2 t}} \, dt \\
&= \int_0^1 \frac{du}{\sqrt{1+u^2}} \\
&= \int_0^{\frac{\pi}{4}} \sec\theta d\theta = \ln|\sec\theta + \tan\theta| \Big|_0^{\frac{\pi}{4}} \\
&= \ln(\sqrt{2}+1).
\end{aligned}$$

$$u = \sin t \quad du = \cos t \, dt$$

$$u = \tan \theta \quad du = \sec^2 \theta \, d\theta$$

Example 7 : $\int \sqrt{\frac{a+cx}{x}} \, dx$

Hint : 令 $x = ?$ 可以將 $\sqrt{\quad}$ 除掉.